Expertise (lecture note)

Lecturer: Benjamin Lipp

Introduction

This lecture introduces the sociology of scientific knowledge and expertise, which understands **science as a primarily social activity** not only a collection of formal methods. Science happens in a social context, for example, in a particular nation state, university or laboratory. It is thus fundamentally a collaborative, socially organised endeavour in the pursuit of producing reliable knowledge. This **social context shapes scientific activity**, how scientists view the world, what they find interesting, and how they approach it. This is inevitable. A philosophical thought experiment to make this point is the **Experimenter's Regress**: a circular dilemma where scientists cannot judge an experiment's validity without assuming the correctness of the outcome, nor assess the outcome without assuming the apparatus works properly. In practice, this is rarely a big issue because scientists instead of only relying on experimental methods trust that their basic assumptions are correct. So, in fact, without these social institutions (also called **paradigms**), science would not work. As there is *no absolute* way of knowing whether knowledge is reliable, we need to *trust* it (for the time being).

This does not mean that the scientific method is inherently wrong or flawed. Instead, recognising the social shaping of science invites us to examine where this uncertainty occurs, how paradigmatic assumptions reduce it and what happens if they cannot, for example, in the case of a controversy about what counts as evidence and what not. In doing so, sociology shifts focus from science as a formal method to science as a social practice organised by particular paradigms. This insight also has implications for the interaction between different experts or other outsiders of a given field. In such contexts (also called **trading zones**) scientists require **interactional expertise**, one which is fluent in other fields' language and thus able to mediate between them. This approach is better suited for navigating complex wicked problems.

Paradigms

Science is often portrayed as a gradual accumulation of facts (aka progress). The historian Thomas Kuhn challenged this view by showing that rather than a linear process, the history of science saw periods of "normal science" punctuated by **paradigm shifts**, i.e., states of crisis in which multiple paradigms compete with one another and eventually settle around a new paradigm.

Normal science refers to research conducted under a prevailing paradigm. People that accept a shared set of assumptions, concepts, and methods form a **thought collective**. What makes a collective can be formal methods or theories, but also more implicit assumptions about how certain problems should be approached or how a given method is specifically used. Members of a thought collective devote themselves to "puzzle-solving": applying established theories to new cases, refining measurements, and filling in details. This is possible because scientists are always part of a wider community that disciplines its members according to implicit norms and established standards. Discrepancies (anomalies) that arise are typically set aside or explained away, because the community trusts the paradigm's overall framework.

A **paradigm shift** occurs when anomalies accumulate to the point that they start to challenge the over-arching paradigm. This is because paradigms are only partial representations of the world. In such a situation, facts are uncertain and the methods by which to produce those facts are in dispute. If an alternative is on the horizon, more scientists will adopt the alternative. Note that this is not simply a question of being rational, rather there are social factors influencing it: for example, the scientists most ready to accept a new paradigm are typically junior scholars who have not built their careers on the beliefs and practices of the older paradigm. Eventually, as older and 'conservative' scientists become less important, that alternative may become a paradigm itself, structuring a new period of normal science.

This shows how paradigms function as a *worldview* (providing categories for interpreting phenomena) and as a *form of life* (structuring laboratory practices, instrument design, and peer-review etiquette).

Example: Take late-twentieth-century neuroscience. Under the neuroanatomical paradigm (mind as the product of localized brain structures and gross lesions), researchers mapped cortical regions, correlated injuries with deficits, and explained behavior through static brain architecture. When persistent anomalies (limits of lesion studies, new imaging showing dynamic neural processes, molecular findings linking genes, neurotransmitters, and circuits) eroded confidence in purely anatomical accounts, the field entered a period of rethinking. The neuromolecular paradigm (mind as emerging from molecular events in neural networks) replaced the older framework, reorganizing both theoretical language (e.g., "memory" as synaptic plasticity) and practical apparatus (from post-mortem dissection to fMRI, PET, and molecular assays). This was not gradual accumulation of knowledge but a paradigm shift: what counted as a valid explanation, a critical experiment, or a "brain disorder" changed abruptly.[1]

Interactional expertise and trading zones

When disciplines with different paradigms interact – e.g., engineering teams working with molecular biologists, policy makers consulting physicists – a new problem arises: how can people communicate if they don't share the specialized practices and worldview of a field? Harry Collins and others use two concepts – **trading zones** and **interactional expertise** – to explain how such cross-disciplinary communication can succeed even in the absence of a shared paradigm (see Further Readings section).

Borrowing the metaphor of economic "trading zones," Collins et al. showed that when two groups have different languages or cultures (e.g., clinicians and computational modelers), they can nevertheless "trade" by developing partial, localized languages—inter-languages—that allow enough mutual understanding to collaborate on a shared goal. Within trading zones, actors negotiate translations of key terms, agree on shared protocols, and gradually build what amounts to a common language specific to the collaboration.

[1] If you want to know more about this, see Abi-Rached JM and Rose N (2010) The birth of the neuromolecular gaze. *History of the Human Sciences* 23(1): 11–36.

A **trading zone** is any space – geographic, institutional, or conceptual – where participants from different disciplines agree on enough common vocabulary and shared practices to collaborate, despite deeper incompatibilities in worldview or method. Some trading zones are cooperative (participants voluntarily negotiate new terms), others are coercive (a dominant culture imposes its terms), and their outcomes vary from fully merged cultures (creoles) to flexible "boundary objects" (shared instruments or datasets around which each group still interprets differently).

To engage in these trading zones experts do not only need expertise in their own field (contributory expertise) but also expertise that relates to the interaction with experts from other disciplines (interactional expertise).

- **Contributory expertise:** Full mastery of both the language and hands-on practices of a field – for example, a molecular biologist who can both talk fluently about and themselves run CRISPR experiments.
- **Interactional expertise:** Fluency in a field's discourse – its jargon, tacit rhetorical moves, and standards – without the ability to carry out the technical work. For example, a science journalist who can discuss nuances in particle-physics debates yet has never set foot in a collider control room.

Interactional experts are valuable boundary-spanners: they can translate concepts across fields, anticipate misunderstandings, and negotiate shared problem definitions without being able to operate the specialized tools themselves.

Conclusion

For the **responsible engineer**, understanding paradigms and interactional expertise is essential in contemporary, interdisciplinary settings—especially in situations of controversy when experts disagree. In such cases, different disciplines apply competing frameworks to the same problem. Recognizing that each discipline operates under its own paradigm – its own set of knowledge practices and assumptions about the world – helps the engineer see why disagreements persist. Instead of assuming one side is simply "wrong," a responsible engineer can identify which anomalies or boundary conditions each paradigm struggles to incorporate and propose solutions that integrate insights from both sides.

In these situations, responsible engineers must develop interactional expertise in adjacent fields. For example, learning to think like a clinician or a pharmacologist (even without practicing medicine) allows the engineer to anticipate usability concerns, safety requirements, and regulatory hurdles. Similarly, forming trading zones – e.g., monthly design workshops where each discipline presents in its own language but aligns on shared project goals – prevents miscommunication and ensures that all languages are honoured.

Further readings

For a general introduction into the topic:

Sismondo S (2010) The Prehistory of Science and Technology Studies. In: *An Introduction to Science and Technology Studies*. Oxford: Blackwell Publishing, pp. 1–11.

For an introduction into the specific concepts discussed above:

For the concept of paradigm: Sismondo S (2010) The Kuhnian Revolution. In: *An Introduction to Science and Technology Studies*. Oxford: Blackwell Publishing, pp. 12-22.

For the concept of interactional expertise and trading zones: Collins H, Evans R and Gorman M (2007) Trading zones and interactional expertise. *Studies in History and Philosophy of Science* Part A 38(4): 657–666.